Validating the jNO PEEC solver

Where the disagreement came from, and what closed it

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Three solvers disagree on the same power module

The measurement. Loop inductance between the DC+ and DC– terminals of one layout, at 1 MHz, on geometry all three codes read from the same file.

	loop inductance	
Ansys Q3D (AC RL)	20.641 nH	21 209 elements, converged to 0.331 %
pypeec 5.8.0	21.4 nH	not converged
jNO	26.4 nH	+28 %

The reference is not the weak point. Q3D's AC RL solve is adaptively refined over twelve passes and independently converged. So a 28 % gap has to be explained on our side — and the first job is to decide whether it is the method or the mesh.

The method is right where an exact answer exists

The case. One straight copper bar, 40 × 4 × 0.51 mm, driven end to end at 1 kHz — the current fills the section, so both the resistance and the inductance are known.

	R	error	L	error
analytic	338.0 $\mu\Omega$	—	27.207 nH[1]	—
jNO	337.2 $\mu\Omega$	−0.2 %	27.731 nH	+1.9 %
pypeec	332.7 $\mu\Omega$	−1.6 %	26.673 nH	−2.0 %

The result. jNO lands on the exact DC resistance and sits inside Grover's own accuracy. Whatever the module gap is, it is not the partial-inductance formulation.

[1] Grover, *Inductance Calculations*, 1946

The gap is resolution, amplified by cancellation

One cell across the narrowest conductor. Every layout has a 1.00 mm minimum trace width, and the runs above used a 1.0 mm pitch. Current crowding at a trace edge cannot appear in a discretisation that has no edge to crowd against.

And the quantity reported magnifies it. The loop inductance with the baseplate is a small difference of two large numbers,

$$20.6 \;=\; 54.5 \;-\; 33.9 \quad \mathrm{nH},$$

so a 1 % error in the bare loop arrives as 2.64 % in the answer. Converging on the plate-less loop instead removes the amplification — and it is the only quantity here with a clean reference sequence.

What that predicts. Uniform refinement should march toward 54.505 nH and be unaffordable long before it arrives. It does: 58.803 → 57.957 → 57.340 nH at 1.0, 0.5, 0.4 mm, i.e. 13 381 → 53 458 → 83 705 elements to remove half the error.

Corrected, the two PEEC codes agree

	plate-less loop	
Ansys Q3D (AC RL)	54.505 nH	converged, 0.331 % over 12 passes
jNO, extrapolated	54.87 nH	+0.7 %
pypeec, extrapolated	55.69 nH	+2.2 %

Two independent codes, one answer. At matched pitch jNO and pypeec agree to ~ 1 %, and both extrapolate onto Q3D's converged value. The +28 % was under-resolution seen through a 2.64× amplifier — not a solver defect in any of the three.

Priced and dismissed along the way. Frequency (flat 1–10 MHz), the ground plane (already included, at the real separation), capacitance (663× the inductive impedance), bond wires, the dies, and lateral mesh density — each measured and eliminated rather than argued away.

Putting the elements where the physics needs them

If the diagnosis is right, grading the mesh should pay for itself. Every trace is an axis-aligned rectangle, so the coordinates to refine toward are axis-aligned — which a graded rectilinear grid expresses exactly, with no hanging nodes. It gives up translation invariance, so the FFT no longer applies and the operator becomes hierarchical (H-matrix + ACA).

mesh	elements	L	vs. 54.505 nH
uniform 1.0 mm	13 381	58.803 nH	+7.88 %
uniform 0.5 mm	53 458	57.957 nH	+6.33 %
uniform 0.4 mm	83 705	57.340 nH	+5.20 %
graded	8 192	57.299 nH	+5.13 %
graded	14 111	56.604 nH	+3.85 %

8 192 graded elements match 83 705 uniform ones, and 14 111 beat them outright while still falling — where the uniform sequence had flattened. The elements were never the problem; their placement was.